Predicting Moment of Learning in ASSISTments: Model Simplicity, Accuracy, and Student Behavioral Differences

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Abstract

The "moment of learning" probability $P(J)$ estimates how likely a student is to have just learned a skill at a given practice opportunity. Published $P(J)$ models use large feature sets. We test whether a model using only correctness and hint use predicts $P(J)$ as well as the full-feature model, and examine how student behavioral profiles relate to learning patterns. On 119,141 labeled ASSISTments attempts (3,622 students, 55 skills), evaluated on held-out students, the reduced model reached $R^2 \approx 0.09$, against 0.21 (ridge) and 0.35 (gradient boosting) for the full model. The simplification therefore costs substantial accuracy. Part of the full model's advantage may come from features that proxy the Bayesian Knowledge Tracing (BKT) state used to build $P(J)$, so it cannot be read as purely behavioral signal. A k-means analysis separated a high-help-seeking group (15%) from a low-help-seeking group (85%). The high-help group had lower mean $P(J)$ but no different spikiness. Within both groups the full model clearly outperformed the reduced one, and we found no evidence that the accuracy gap differs between groups, although the smaller group is underpowered.

1. Introduction and Literature Review

Intelligent tutoring systems log every attempt, hint request and response time, yet knowing *when* learning happens is hard. Bayesian Knowledge Tracing (BKT) models a latent binary knowledge state per skill, updated from correctness with guess, slip and learn parameters (Corbett & Anderson, 1995). Baker et al. (2011) extended it to estimate the probability that learning occurred *at a specific step*, $P(J)$. This was done by combining BKT estimates with a model fitted to labels derived from later performance. Plotting $P(J)$ over time often shows a "spiky" pattern, where one or two moments carry most of the learning, and the spikiness predicts robust learning and preparation for future learning (Baker et al., 2013). Related work links help-seeking and hint use to learning and to unproductive behavior such as gaming the system (Aleven & Koedinger, 2000; Baker et al., 2004). The data come from ASSISTments, a widely used online math platform (Heffernan & Heffernan, 2014).

Reading this literature, we identified three gaps. First, $P(J)$ models are typically built from large, hand-engineered feature sets, mixing performance, timing and history features. We found no systematic test of how much predictive power remains when only the two most universally logged signals, correctness and hint use, are available. Such a test matters for platforms with limited logging. Second, studies usually report one pooled accuracy figure, but learners differ in help-seeking and pacing, so a simplified model may fail unevenly across groups. Third, rich history features such as cumulative wrong counts overlap with the BKT quantities that define the label. Comparisons of feature sets therefore risk crediting a model for "behavior" that is partly the label's own construction. We found these gaps by comparing what prior studies report (feature sets, pooled metrics, label construction) with what is needed to judge a simplified model fairly (feature ablation, subgroup evaluation, leakage screening).

Hypotheses

- **H1 (temporal behavior).** Hint rate, error rate and time taken change across successive practice opportunities on a skill (a decrease is expected). We compute each measure by opportunity number (1–10), overall and by profile, using medians for skewed times. The evidence is descriptive and cross-sectional. A measure is supported only if it moves consistently

by a meaningful amount; a single drop followed by a plateau counts as weak support. No p-values are reported without a formal test.

- **H2 (profile differences).** Profiles differ in temporal patterns and in $P(J)$ distributions. We compare student-level mean $P(J)$ and spikiness (maximum divided by mean $P(J)$ per student–skill sequence) using two-sided Mann–Whitney U tests, rank-biserial effect sizes and Holm-corrected p-values. Support requires an adjusted $p < 0.05$ with a non-negligible effect. Only measures not used for clustering count as independent evidence.
- **H3 (reduced vs. full model).** A correctness-and-hint model predicts $P(J)$ with accuracy comparable to the full model, though possibly lower. Both models are fitted on the same target and evaluated on held-out students (80/20 split plus five-fold grouped cross-validation) using R^2, MAE and RMSE. A student-level bootstrap gives intervals for the difference. No equivalence margin was pre-specified, so a nonsignificant difference is not read as equivalence.
- **H4 (gap across profiles).** The accuracy loss from simplification differs between profiles. We evaluate pooled held-out predictions within each profile and define the gap as $R^2_{\text{full}} - R^2_{\text{reduced}}$. A student-level bootstrap (500 resamples) gives intervals for each gap and for their difference. H4 is supported only if the interval for the difference excludes zero; otherwise we report no evidence of a difference, not equality.

2. Methodology

Data. We used the ASSISTments $P(J)$ feature-and-label file: 187,312 attempt-level rows, 4,139 students and 55 skills. Fields include student and skill identifiers, timestamps, correctness, hint use, attempt counts, response times, cumulative history features, BKT estimates (prior/posterior knowledge, G, S, T) and $P(J)$ as the target. Cleaning removed invalid values (4 negative response times set to missing). No duplicates or malformed rows were found. In total, 119,141 rows have a valid $P(J)$ label and 68,171 contain the placeholder "." and are unlabeled. Regression uses labeled rows only.

Descriptives and temporal analysis (RQ1). For each practice opportunity number, we computed mean hint rate, error rate and time taken, and plotted $P(J)$ by opportunity and by correctness–hint cell. Temporal curves are also reported separately for the two behavioral profiles for opportunities 1–10. These are cross-sectional comparisons by opportunity number rather than within-student trajectories, because students with longer sequences are a selected subset.

Behavioral profiles (RQ2). Student-level aggregates were computed for students with at least 20 rows ($n = 2{,}107$). Features were z-scored and clustered with k-means (MacQueen, 1967), and k was chosen by silhouette width (Rousseeuw, 1987). Clustering used seven standardized student-level features: first-response correct rate, hint rate, help-request rate, bottom-out-hint rate, median log response time, share of fast responses (at or below the 10th percentile of all response times), and mean sequence length per skill. Final L_n was not used, and $P(J)$ was held out as an external check only. No tertile-split fallback was run.

Profile differences were assessed using two-sided Mann–Whitney U tests and rank-biserial effect sizes, with Holm correction across measures. Spikiness is the per student–skill ratio of maximum to mean $P(J)$ over labeled rows (at least two labeled rows, positive mean), averaged within student. The comparison of mean $P(J)$ and spikiness provides an external check because neither measure was used to form the clusters. In contrast, differences in hint, help-request, correctness and timing measures are partly circular: these variables contributed directly to cluster formation.

For H4, the pooled models' held-out predictions were evaluated within each profile (students in the 20% test split with at least 20 rows). Models were not refit separately within profiles. Student-level bootstrap (500 resamples) gives confidence intervals for ΔR^2 within each profile and for the difference in ΔR^2 between profiles. These comparisons evaluate whether the pooled models perform differently across the observed groups; they do not establish that profile-specific models would perform better.

Model comparison (RQ3). Three feature sets were defined: (i) *reduced*: correctness and hint only (2 features); (ii) *full*: all 54 allowed features; (iii) *full minus high-risk*: the full set without 22 features computed from post-first-response behavior, which are the most likely to carry label information. $P(J)$, BKT states and parameters, IDs, timestamps, and opportunity/sequence-length counts that use future rows were excluded from predictors. Each set was fitted with ridge regression (Hoerl & Kennard, 1970) and histogram-based gradient boosting (HGB) from scikit-learn (Pedregosa et al., 2011), with fixed default hyper-parameters. Medians for imputation were learned on training data inside pipelines. Data were split 80/20 *by student* (2,897 train / 725 test students, no overlap), and we also ran 5-fold student-grouped cross-validation. We report R^2, MAE and RMSE on held-out students, and a 95% student-level bootstrap CI (Efron & Tibshirani, 1993) for the full-minus-reduced difference. Permutation importance (Breiman, 2001) and a name- and correlation-based leakage screen addressed the third gap.

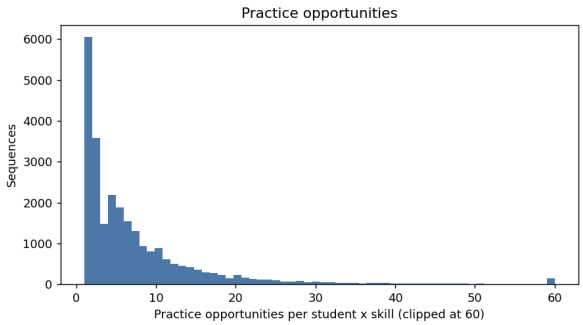
Tools. Python with pandas (McKinney, 2010), NumPy (Harris et al., 2020), SciPy (Virtanen et al., 2020), scikit-learn and Matplotlib (Hunter, 2007).

3. Results

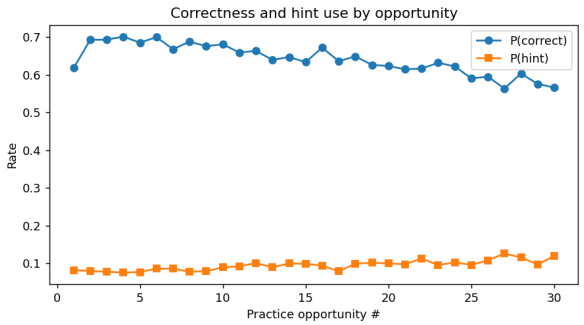
Data and label structure. First-response correctness was 0.657 and hint rate was 0.088. Students had a median of 4 practice opportunities per skill (mean 7.2, max 141). $P(J)$ is highly right-skewed (Table 1): mean 0.0028, median 0.0001, SD 0.0080, 93.1% of values below 0.01, skewness 4.62. Labeled and unlabeled rows differ in standardized terms on correctness (-0.57), opportunity number ($+0.31$), sequence length ($+0.73$) and an original-problem indicator (-0.40), so missingness is not random and conclusions apply to labeled rows only.

Table 1: Mean $P(J)$ by first-response outcome.

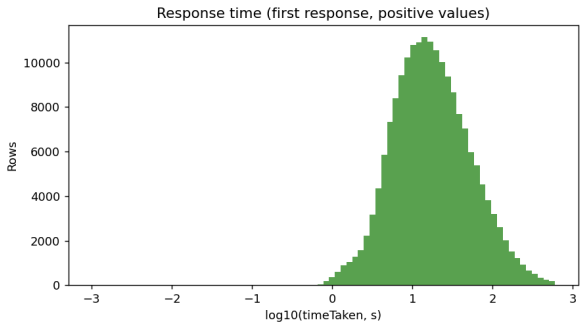
Correct / Hint	n	Mean $P(J)$
Incorrect, no hint	47,828	0.0061
Incorrect, hint	16,488	0.0038
Correct, no hint	122,996	0.0007



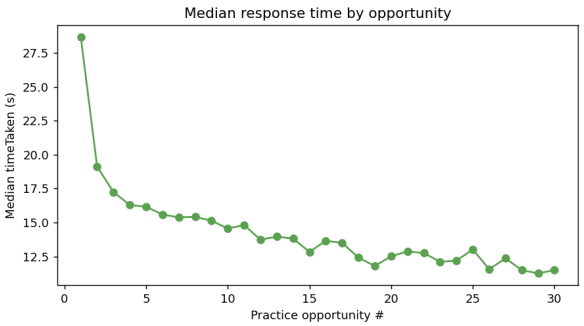
(a) Opportunities



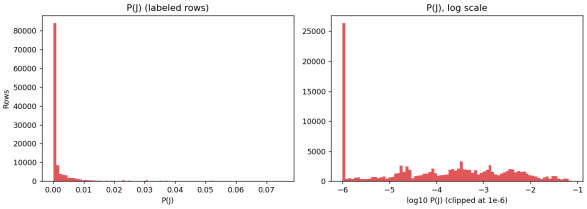
(b) Correct/hint by opp.



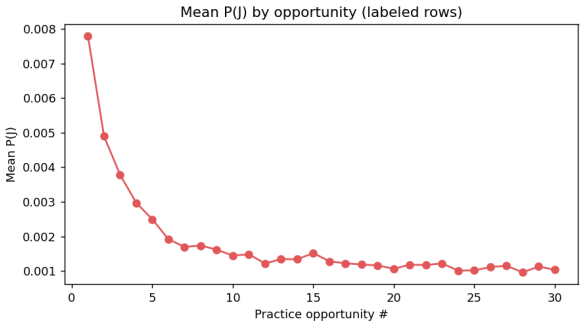
(a) Response time



(b) Time by opp.



(a) $P(J)$ distribution



(b) $P(J)$ by opp.

Figure 3: Descriptive patterns: practice opportunities per student–skill, correctness and hint use, response time, and $P(J)$ across opportunity numbers.

RQ1: temporal patterns. Table 2 and Figure 4 show hint rate, error rate and median time taken for opportunities 1–10 by profile. Time taken falls in both profiles (P0: 28.1 to 15.4; P1: 20.9 to 7.8, as logged in `timeTaken`), with most of the drop occurring by opportunity 4. Error rate in P0 falls from 0.360 to 0.281 between the first and second opportunity and then stays near 0.27–0.29. In P1 it does not fall (0.663 at opportunity 1, 0.662 at opportunity 10). Hint rate in P0 falls from 0.061 to 0.043 by opportunity 5 and then drifts back to 0.051. In P1 it rises from 0.281 to 0.366. H1 is therefore supported for time taken in both profiles, only weakly supported for error rate in P0 (a one-step drop followed by a plateau), and not supported for hint rate or error rate in P1. These curves are cross-sectional by opportunity number. Students with long sequences are a selected subset, so later points do not track the same students.

Table 2: Hint rate, error rate and median time taken at selected opportunities, by profile.

	Hint rate		Error rate		Median time	
Opp.	P0	P1	P0	P1	P0	P1
1	0.061	0.281	0.360	0.663	28.1	20.9
2	0.053	0.334	0.281	0.607	19.2	12.4
5	0.043	0.318	0.278	0.636	17.0	9.5
10	0.051	0.366	0.276	0.662	15.4	7.8

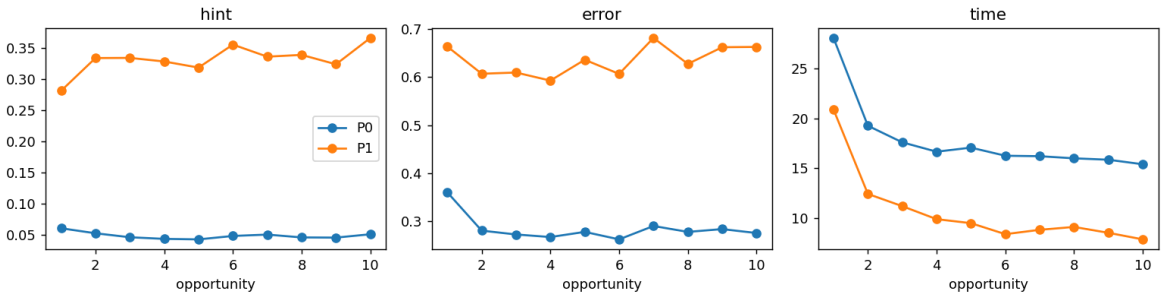
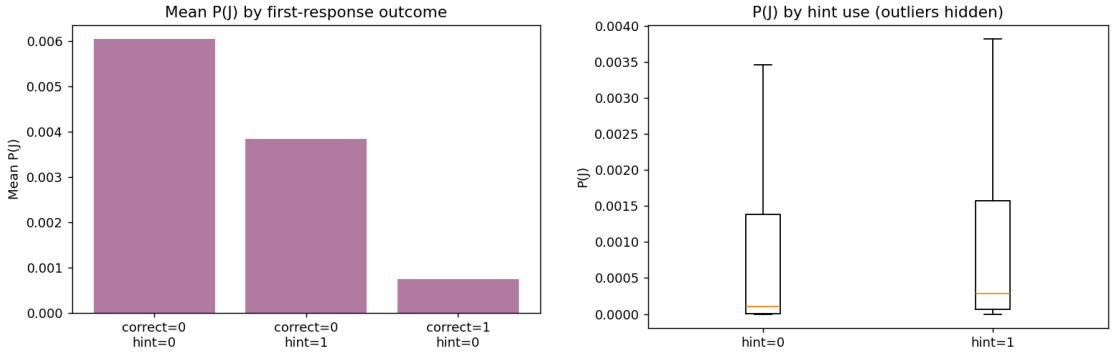


Figure 4: Hint rate, error rate and median time taken by opportunity number (1–10) and profile.

Table 1 and Figure 5 show that $P(J)$ is highest after incorrect, no-hint responses and lowest after correct ones. This is consistent with $P(J)$ concentrating on moments after errors, when a learning event is most informative.



(a) $P(J)$ by correct/hint **(b)** $P(J)$ by hint use

Figure 5: $P(J)$ by first-response outcome.

RQ2: behavioral profiles. Silhouette selection chose $k = 2$ (silhouette 0.526) among the five candidate values (Table 4). Profile P0 ($n = 1{,}794$) shows low help-request and hint rates. Profile P1 ($n = 313$) shows high help-request and hint rates. Mean $P(J)$ of members is higher for P0 (0.0033) than for P1 (0.0021). The profiles are exploratory and unvalidated.

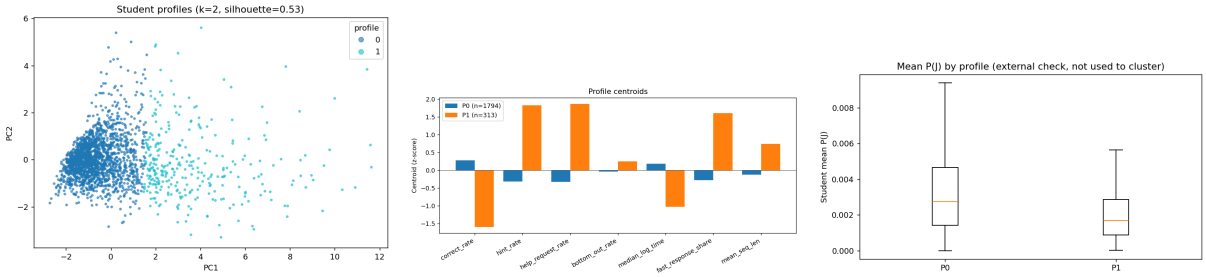
Table 3 gives two-sided Mann–Whitney U tests at the student level, with rank-biserial effect sizes and Holm-corrected p-values. P1 students have much higher help-request, hint and error rates, answer faster, and have longer skill sequences (all Holm-adjusted $p < .0001$). Because these measures were used to form the clusters, such differences are expected by construction and are not independent evidence. The two measures not used in clustering are more informative. Mean $P(J)$ is lower in P1 (median of student means 0.0017 vs. 0.0027, rank-biserial $r = 0.28$, $p < .0001$). Spikiness does not differ (medians 5.07 vs. 4.59, $r = -0.01$, $p = .71$). H2 is supported for temporal patterns and mean $P(J)$, but not for spikiness.

Table 3: Mann–Whitney U tests, P0 ($n = 1{,}794$; spikiness $n = 1{,}787$) vs. P1 ($n = 313$). r is the rank-biserial correlation (positive: P0 higher).

Measure	Median P0	Median P1	U	r	Holm p
Help-request rate	0.033	0.365	15,892	−0.94	$< .0001$
Hint rate	0.028	0.317	21,507	−0.92	$< .0001$
Error rate	0.281	0.648	22,249	−0.92	$< .0001$
Median log time	2.93	2.33	463,695	0.65	$< .0001$
Mean sequence length	7.12	12.50	151,847	−0.46	$< .0001$
Mean $P(J)$	0.0027	0.0017	360,149	0.28	$< .0001$
Spikiness	4.59	5.07	275,991	−0.01	.71

Table 4: Behavioral profiles (k-means, $k = 2$).

Profile	Description	n students	Mean $P(J)$
P0	Low help-request, low hint rate	1,794	0.0033
P1	High help-request, high hint rate	313	0.0021



(a) PCA of profiles **(b)** Centroids (z) **(c)** $P(J)$ by profile

Figure 6: Behavioral profiles from k-means and their $P(J)$ distributions.

RQ3: reduced vs. full model. Table 5 gives held-out-student metrics. Predicting the training mean yields $R^2 \approx 0$. The reduced model reaches $R^2 \approx 0.088$ with either learner, and ridge and HGB are equivalent because two binary features carry little nonlinear structure. The full model reaches 0.2078 (ridge) and 0.3539 (HGB), with train/test R^2 of 0.425/0.354 for HGB, indicating moderate overfitting. Removing the 22 high-risk features changes HGB only slightly (0.3481) and ridge somewhat more (0.1854).

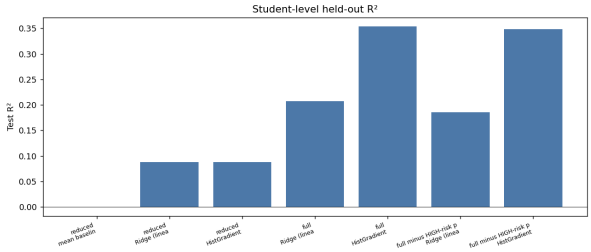
Table 5: Test performance on held-out students (725 students, 26,171 rows).

Feature set	Model	R^2	MAE	RMSE	CV R^2
Mean baseline	—	−0.0001	0.00401	0.00783	—
Reduced (2)	Ridge	0.0876	0.00370	0.00748	0.095
Reduced (2)	HGB	0.0880	0.00372	0.00747	0.094
Full (54)	Ridge	0.2078	0.00365	0.00697	0.206
Full (54)	HGB	0.3539	0.00283	0.00629	0.352
Full minus high-risk (32)	Ridge	0.1854	0.00375	0.00706	0.184
Full minus high-risk (32)	HGB	0.3481	0.00284	0.00632	0.342

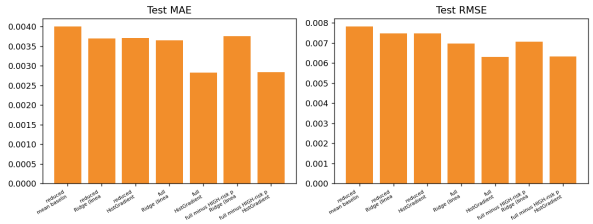
Student-level bootstrap (Table 6) shows that the gain in R^2 from the full model is clearly nonzero for both learners (CIs exclude 0). For MAE, the ridge improvement is not distinguishable from zero, while the HGB improvement is.

Table 6: Full minus reduced: test difference with 95% student-bootstrap CI.

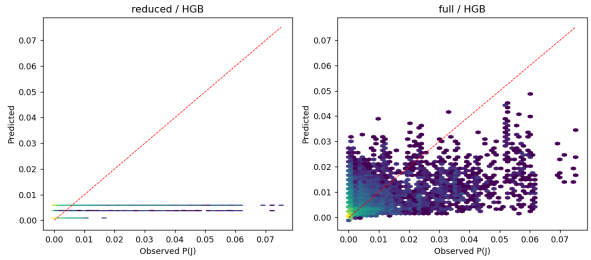
Model	ΔR^2 [95% CI]	MAE reduction [95% CI]
Ridge	+0.1202 [0.1116, 0.1286]	+0.00005 [−0.00001, 0.00011]
HGB	+0.2659 [0.2428, 0.2872]	+0.00089 [0.00082, 0.00096]



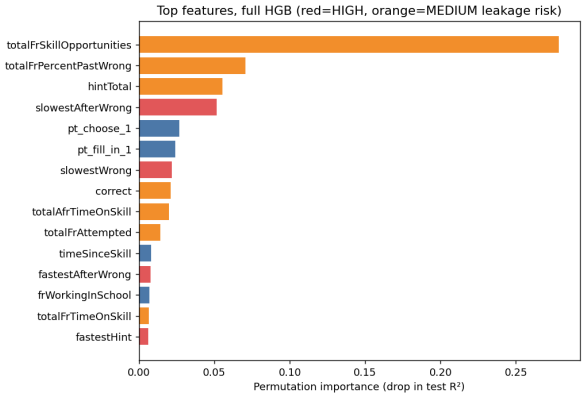
(a) Test R^2



(b) MAE/RMSE



(c) Pred. vs obs.



(d) Importance

Figure 7: Model comparison on held-out students.

Leakage screen and importance. The top permutation importances for the full HGB are `totalFrSkillOpportunities` (0.2785), `totalFrPercentPastWrong` (0.0705), `hintTotal` (0.0553), `slowestAfterWrong` (0.0515) and `pt_choose_1` (0.0268). The largest contributor, a practice-opportunity count, is tied to label missingness by sequence position, so it may reflect label construction rather than behavior. Only `correct` exceeded $|r| > 0.5$ with $P(J)$, L_n or L_{n-1}.

H4: gap across profiles. We evaluated the pooled models' held-out predictions separately within each profile (P0: 375 test students, 20,483 rows; P1: 54 students, 3,966 rows). Table 7 shows that the full model beats the reduced model in both profiles. For HGB, ΔR^2 is 0.266 in P0 (95% CI 0.242–0.290) and 0.241 in P1 (0.142–0.350). For ridge it is 0.116 (0.106–0.126) and 0.154 (0.088–0.249). The difference in gaps between profiles is not distinguishable from zero (HGB: +0.024, CI −0.082 to 0.123; ridge: −0.040, CI −0.134 to 0.030). H4 is therefore not supported. Absolute accuracy is lower in P1 at both feature levels. With HGB, correlation with true $P(J)$ is 0.32 (reduced) and 0.61 (full) in P0, against 0.15 and 0.37 in P1. The reduced model's R^2 in P1 is negative (−0.17 for HGB), meaning it predicts worse than that group's own mean. This likely reflects miscalibration of a pooled model on a group with a different $P(J)$ level, which we did not test.

Table 7: Reduced vs. full model within profile (held-out students). r is Pearson correlation with true $P(J)$.

Model	Profile	R^2_{red}	R^2_{full}	r_{red}	r_{full}	ΔR^2	95% CI
Ridge	P0	0.102	0.218	0.319	0.467	0.116	[0.106, 0.126]
Ridge	P1	−0.192	−0.039	0.154	0.282	0.154	[0.088, 0.249]
HGB	P0	0.101	0.367	0.319	0.607	0.266	[0.242, 0.290]
HGB	P1	−0.168	0.073	0.154	0.372	0.241	[0.142, 0.350]

4. Discussion and Limitations

Interpretation. For H3, the data reject equal accuracy. With only correctness and hint use, the model explains about 9% of variance in $P(J)$, and the full HGB explains about 35%, a gain of

0.27 R^2 (CI 0.24–0.29). The reduced model does capture the main directional pattern (highest $P(J)$ after unaided errors), so it is a usable coarse signal but not a replacement. However, the most informative features are cumulative-history counts, which approximate the BKT state and the sequence position behind the label. The gap therefore partly measures how much of BKT's own information the full model can recover, and it should not be read as evidence that richer behavior causes learning. The small change when dropping high-risk features suggests the gain is not driven only by post-response timing features, but it does not rule out leakage through the remaining history features.

For H1, time taken falls in both profiles, but error and hint rates do not fall consistently. The only clear improvement is a one-step drop in P0 error rate, and P1 shows no improvement and a rising hint rate. This suggests that simple practice-opportunity curves hide large differences between students. For H2, the profiles separate on help-seeking, error rate and speed by construction, and the independent evidence is that the high-help group has lower mean $P(J)$ but equal spikiness. High-help students therefore do not show more or less concentrated learning moments, only lower average $P(J)$. For H4, the reduced model gains from added features in both profiles by similar amounts, and the confidence interval of the difference includes zero. We therefore find no evidence that simplification costs one profile more than the other. Given 54 P1 test students, this is absence of evidence rather than evidence of equality. Both models do worse in P1 in absolute terms, so a correctness+hints-only model is a particularly weak proxy for heavy help-seekers.

Limitations.

- *Label construction.* $P(J)$ construction is undocumented in the files provided, so leakage cannot be proven present or absent. Name-based risk flags are heuristics.
- *Non-random missingness.* 36% of rows are unlabeled, and these differ systematically from labeled rows (shorter sequences, lower correctness). Results do not generalize to unlabeled rows.
- *Skewed target.* 93% of $P(J)$ values are below 0.01, so R^2 and MAE are driven by a small number of high-value rows, and absolute accuracy values are small.
- *Profiles.* Clusters are exploratory, based on 2,107 students with at least 20 rows, and are unvalidated. The $k = 2$ solution is unbalanced (15% vs. 85%), and students with shorter histories are excluded.
- *Modeling.* A single 80/20 split plus grouped CV is used, with untuned default hyper-parameters. Results are observational, with no causal claims.
- *Circular profile differences.* Clusters were formed on hint, help-request, correctness and timing features, so group differences on those measures are expected by construction. Only mean $P(J)$ and spikiness are independent checks.
- *Small test group and pooled models.* H4 uses 54 P1 test students, and the models were trained on pooled data rather than refit per profile, so the gap comparison is low-powered and the miscalibration in P1 is untested.

Practical implication. Platforms logging only correctness and hints can obtain a coarse $P(J)$ proxy, but should not expect parity with full-feature models, and any reported gains from rich features should be checked against how the label was built.

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5. Code Availability and Reproducibility

All programmatic workflows, including data cleaning, exploratory data analysis, feature engineering, behavioral clustering, hypothesis testing, and model evaluation, were implemented in Python. The analysis uses pandas and NumPy for data manipulation, SciPy for statistical testing, scikit-learn for machine learning and clustering, and Matplotlib for visualization.

The Jupyter notebook, `ET610_Project.ipynb`, contains the analysis workflow, including model comparisons, behavioral profile analysis, and evaluation of predictive performance. Generated figures and tables are organized in the `outputs/figures/` and `outputs/tables/` directories, respectively. The accompanying report documents the methodology, results, and limitations.

The notebook is available for independent inspection and reproduction. Reproducing the results requires access to the ASSISTments dataset and configuration of the corresponding data paths in the notebook.

GitHub Repository: `https://github.com/VanshikaNalamasa/ET-610-Project`